# Loïc Guezou

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## Current position

**PhD student**, Université Paris-Saclay, ENS Paris-Saclay, CentraleSupélec, CNRS – Laboratoire de Mécanique Paris-Saclay (LMPS)
**Thesis**: *Development of hyperelastic reduced-order models using Proper Generalised Decomposition (working title)*
**Defense date**: mid-2026
**Jury**: TBD

## Research interests

Computational Mechanics; Model Order Reduction; Hyperelasticity; Biomechanics

## Education

**Ph.D. in Computational Mechanics**
Université Paris-Saclay — 2022 –

**M.Sc. in Computational Mechanics**
Université Paris-Saclay — 2021 – 2022

**M.Sc. in Mechanical Engineering**
ISAE-Supméca — 2019 – 2022

**CPGE (preparatory cycle to the top Engineering schools entrance exams)**
Lycée Condorcet, Paris — 2017 – 2019

## Experience

**Ph.D. student in Computational Mechanics** — Gif-sur-Yvette, France
Laboratoire de Mécanique Paris-Saclay — October 2022 - April 2026

Skills developed: Model order reduction (PGD), Hyperelasticity theory, Python, Finite Element Method, Numerical methods, Biomimetic Engineering, Scientific writing

**Teaching** — Gif-sur-Yvette, France
CentraleSupélec — 2023 – 2025 (24h)

Teaching assistant for Computational Structural Dynamics. Master 2 students. 2 years in a row.

**Teaching** — Gif-sur-Yvette, France
CentraleSupélec — 2022 – 2024 (40h)

Project supervision on Mechanical Engineering for 3D printing. Master 1 Engineering students. 2 years in a row.

**Research intern in Computational Mechanics** — Gif-sur-Yvette, France
Laboratoire de Mécanique Paris-Saclay — April 2022 - September 2022

Master research internship. Developing a PGD dual formulation (pressure-velocity) for acoustic dynamic computations. Supervision: B. Tie, A. Barbarulo

**Engineer assistant intern in Research & Development** — Morestel, France
NUVIA Protection — September 2020 - January 2021

Quantification of the head loss in fire protection devices for the nuclear sector.

**Student intern** — Paris, France
Laboratoire de Physique de l'Ecole Normale Supérieure — January 2020

## Publications

1. Loïc Guezou, Andrea Barbarulo, Bing Tie. A new PGD parametric modelling strategy for hyperelastic materials. *In the works.*
2. Loïc Guezou, Andrea Barbarulo, Bing Tie. Development of high-fidelity numerical models for 3D printing of patient-specific anatomy. *ECCOMAS Congress 2024 - The 9th European Congress on Computational Methods in Applied Sciences and Engineering*, Lisbon, Portugal; Abstract.
3. Loïc Guezou, Andrea Barbarulo, Bing Tie. Development of high-fidelity numerical models for 3D printing of patient-specific anatomy. *16ème Colloque National en Calcul de Structures (CSMA 2024)*, Hyères, France; Abstract.

## Talks

1. Loïc Guezou, Andrea Barbarulo, Bing Tie. Poster presentation. *7th International Workshop on Model Order Reduction Techniques*, 2025 November 26; Zaragoza, Spain; Poster.
2. Loïc Guezou, Andrea Barbarulo, Bing Tie. Presentation. *9th European Congress on Computational Methods in Applied Sciences and Engineering*, 2024 June 07; Lisbon, Portugal; Talk.
3. Loïc Guezou, Andrea Barbarulo, Bing Tie. Poster presentation. *16th Congress of the Computational Structural Mechanics Association*, 2024 May 14; Giens, France; Poster.

## Skills

**Languages:** English (fluent) | French (native) | Spanish (intermediate) | Italian (intermediate)
**Programming tools:** Python | MatLab | C++ (notions)
**Scientific Software:** CATIA | SolidWorks
**OS:** macOS | Linux | Windows
**Other software:** LaTeX | Typst | Inkscape | Microsoft Office

## Service

ED SMEMaG, Elected representative of the PhD students in the doctoral school council 2022–2024